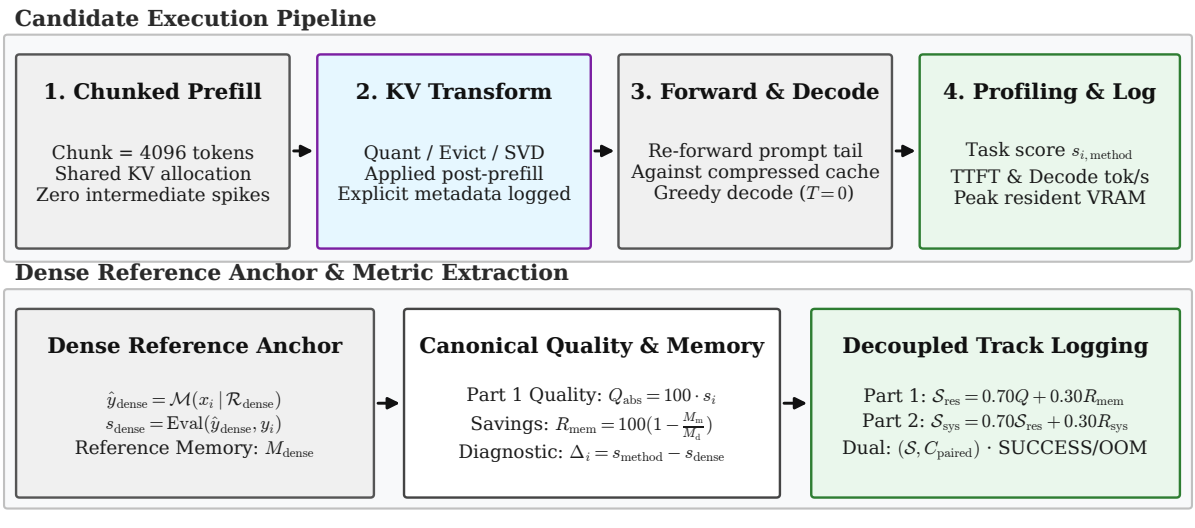
Candidate Execution Pipeline
1. Chunked Prefill
Chunk = 4096 tokens
Shared KV allocation
Zero intermediate spikes
2. KV Transform
Quant / Evict / SVD
Applied post-prefill
Explicit metadata logged
3. Forward & Decode
Re-forward prompt tail
Against compressed cache
Greedy decode ($T=0$)
4. Profiling & Log
Task score $s_{i,\text{method}}$
TTFT & Decode tok/s
Peak resident VRAM
Dense Reference Anchor & Metric Extraction
Dense Reference Anchor
$\hat{y}_{\text{dense}} = \mathcal{M}(x_i \mid \mathcal{R}_{\text{dense}})$
$s_{\text{dense}} = \text{Eval}(\hat{y}_{\text{dense}}, y_i)$
Reference Memory: M_{dense}
Canonical Quality & Memory
Part 1 Quality: $Q_{\text{abs}} = 100 \cdot s_i$
Savings: $R_{\text{mem}} = 100(1 - \frac{M_m}{M_d})$
Diagnostic: $\Delta_i = s_{\text{method}} - s_{\text{dense}}$
Decoupled Track Logging
Part 1: $\mathcal{S}_{\text{res}} = 0.70Q + 0.30R_{\text{mem}}$
Part 2: $\mathcal{S}_{\text{sys}} = 0.70\mathcal{S}_{\text{res}} + 0.30R_{\text{sys}}$
Dual: $(\mathcal{S}, C_{\text{paired}})$ · SUCCESS/OOM